Lab Report : Segmetation the Infection of COVID-19

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I. Introduction

This lab work focuses on applying image segmentation techniques to identify COVID-19 infection in lung X-ray images. The dataset consists of annotated X-ray images with lung and infection masks, enabling both detection and quantification of infected regions.

II. Data Description

The provided dataset contains multiple files; however, this lab work focuses only on the Infection Segmentation Data. The dataset is already organized into three subsets: training, validation, and testing. Each subset is further divided into three categories of chest X-ray images: COVID-19, Non-COVID, and Normal cases.

The training set includes 1,864 COVID-19 images, 932 Non-COVID images, and 932 Normal images. The validation set contains 466 COVID-19 images, 233 Non-COVID images, and 233 Normal images, while the test set consists of 583 COVID-19 images, 292 Non-COVID images, and 291 Normal images. This structured split allows the model to be trained, tuned, and evaluated effectively while maintaining a balanced representation of each class.

III. Methodology and Implementation

A. Data Preparation

The dataset is already divided into training, validation, and testing sets. For this study, only the infection masks were used as ground truth labels for segmentation. Each X-ray image was resized to a fixed resolution to ensure consistent input size for the model.

B. Dataset Organization

To use the dataset with YOLOv8, the data was reorganized into separate `images` and `labels` folders. Each X-ray image has a matching label file that contains the segmentation information for the infection area. In addition, a `data.yaml` file was created to specify the dataset locations, class names, and the total number of classes. This configuration file is required by YOLOv8 for training and validation.

C. YOLOv8 Segmentation Model

I used a YOLOv8 segmentation model because it can perform object detection and segmentation at the same time, while still being fast and efficient. Compared to traditional segmentation models like U-Net, YOLOv8 trains faster and provides good performance, which makes it suitable for this project. The model includes:

- **One-Stage Model:** YOLOv8 performs detection and segmentation in one step, which makes training and prediction faster.
- **Segmentation Output:** The model predicts pixel-level masks that show the infected regions in lung X-ray images.
- **Pre-trained Weights:** The model starts with pre-trained weights so it already understands basic image features.
- **Transfer Learning:** This helps the model learn the COVID-19 infection patterns more easily, even with a limited dataset.

D. Model Training

The model was trained using GPU acceleration on the Kaggle platform to reduce training time. Several training parameters were configured to improve performance and generalization:

- **Epochs:** 80, allowing the model sufficient time to learn infection patterns.
- **Batch Size:** 16, chosen based on GPU memory constraints.
- **Optimizer:** AdamW, which provides stable and efficient weight updates.
- **Data Augmentation:** Techniques such as horizontal flipping and mosaic augmentation were applied to increase data diversity and reduce overfitting.

During training, the model was evaluated on the validation set after each epoch. The best-performing model was automatically saved based on the highest segmentation performance.

E. Evaluation and Results

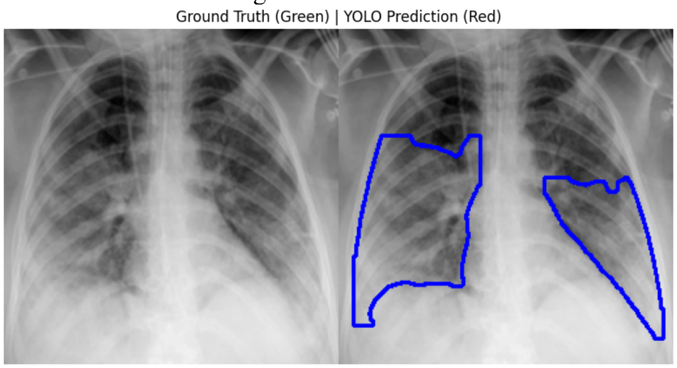
Model performance was evaluated using standard segmentation metrics, including precision, recall, and mean Average Precision (mAP). The primary evaluation metric used in this project is **mask mAP50–95**, which measures segmentation accuracy across multiple Intersection-over-Union (IoU) thresholds. During training, the model showed a stable learning behavior, with most loss values decreasing gradually over time. In the early epochs, the segmentation loss and classification loss were relatively high, indicating that the model was still learning basic infection patterns. As training progressed, these losses consistently decreased, showing improved feature learning and segmentation accuracy.

At the end of training, the model achieved the following overall results:

- **Box mAP@50:** approximately 0.79
- **Box mAP@50–95:** approximately 0.54

- **Mask mAP@50:** approximately 0.78
- **Mask mAP@50–95:** approximately 0.49

This image shows the comparison between the ground-truth infection mask and the model's prediction. The highlighted areas represent infected regions in the lungs. The model is able to identify most of the infection areas, showing that it performs well on this image.



Overall, the model shows good performance in segmenting COVID-19 infection regions. Mask-based metrics are important because they reflect pixel-level segmentation accuracy instead of simple object detection.

IV. Discussion

Although the performance changes slightly between epochs, the overall results improve steadily and eventually stabilize. This is expected because data augmentation is used and medical images are complex. The final results show that YOLOv8 is an effective model for COVID-19 infection segmentation, offering a good balance between accuracy and training speed.